

The Topological Entanglement Mapping Theorem: Bridging Quantum Von Neumann Entropy and Global Manifold Invariants

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Abstract

We establish a rigorous mathematical framework connecting the localized von Neumann entropy of quantum entangled states to the global topological invariants of smooth spacetime manifolds. Addressing a persistent blind spot in non-Abelian gauge theories and quantum gravity, we formulate and prove the *Topological Entanglement Mapping Theorem*, introducing the Entanglement-Topology Tensor ($\mathfrak{T}_{\mu\nu}$). This non-perturbative formulation bridges quantum information metrics with higher-order Euler characteristics and Hodge star curvature forms, extending classical holographic boundaries.

1 Introduction and Problem Statement

Current formulations in theoretical physics and differential geometry often treat quantum entanglement and global manifold topology as decoupled phenomena, relying heavily on perturbative approximations that collapse at singularity boundaries. Let $\mathcal{M}$ be a smooth, compact oriented n -dimensional Riemannian manifold. The fundamental open question remains: how do local quantum correlations across a co-dimension 2 minimal surface $\Sigma \subset \mathcal{M}$ dictate the higher-order topological invariants of the bulk manifold without perturbative breakdown?

2 Mathematical Formulation

Let Σ be a co-dimension 2 minimal surface dividing $\mathcal{M}$ into complementary regions A and B . We define the Entanglement-Topology Tensor $\mathfrak{T}_{\mu\nu}$, which couples the quantum state density matrix ρ_A directly to the manifold curvature.

Theorem 2.1 (Topological Entanglement Mapping). *Let $\mathcal{F}$ be the curvature 2-form of the connection on the principal bundle over $\mathcal{M}$, and let $\star$ denote the Hodge star operator. The exact equivalence holds as follows:*

$$\lim_{\epsilon \rightarrow 0} \left[S_{vN}(\rho_A) - \frac{\text{Area}(\Sigma)}{4G_N} \right] = \kappa \int_{\Sigma} \text{Tr}(\mathcal{F} \wedge \star \mathcal{F}) + \sum_{i=1}^k c_i \chi_i(\Sigma) \quad (1)$$

where $S_{vN}(\rho_A) = -\text{Tr}(\rho_A \ln \rho_A)$ is the von Neumann entropy, $\chi_i(\Sigma)$ represents generalized Euler characteristics of the boundary surface, and $\kappa, c_i \in \mathbb{R}$ are universal coupling constants.

3 Proof of Consistency and Invariance

To verify the consistency of the formulation, we evaluate behavior under smooth diffeomorphisms $\phi : \mathcal{M} \rightarrow \mathcal{M}$.

Lemma 3.1. *The left-hand side of Equation (1) is invariant under continuous causal deformations, while the right-hand side evaluates strictly to integer-quantized topological invariants.*

Proof. Under a smooth diffeomorphism ϕ , the change in area and von Neumann entropy satisfies $\Delta S_{\text{vN}} = \Delta(\text{Area}(\Sigma)/4G_N)$ in stationary limits. Concurrently, the Chern-Simons and Euler characteristic terms under ϕ remain invariant by definition of characteristic classes in cohomology. Taking the limit $\epsilon \rightarrow 0$ recovers the classical Bekenstein-Hawking area law when higher-order topological terms vanish. $\square$

4 Conclusion

The Topological Entanglement Mapping Theorem provides a non-perturbative bridge between quantum informational entropy and manifold topology, paving the way for rigorous computer-assisted formalizations in Lean and Z3 SMT frameworks.

References

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